

Introduction

The scalar curvature of a Riemannian manifold (M, g) is a smooth function $\text{scal}: M \rightarrow \mathbb{R}$. Roughly speaking, the value $\text{scal}_g(x)$ at a point $x \in M$ measures how much the volume of a small geodesic ball around x deviates from that of a Euclidean ball of the same radius. While scalar curvature is the weakest of the standard curvature conditions for a Riemannian manifold, the presence of *positive* scalar curvature can still have strong topological implications. The archetypal example is the celebrated Gauß-Bonnet Theorem:

Theorem (Gauß-Bonnet). *Let (M, g) be a closed Riemannian 2-manifold. Then*

$$\int_M \text{scal} \, dA = 4\pi\chi(M),$$

where $\chi(M)$ is the Euler characteristic of M .

So any closed surface carrying a metric of positive scalar curvature must be diffeomorphic to either the sphere S^2 or $\mathbb{R}P^2$. For non-compact complete manifolds, positive scalar curvature alone may be too weak a condition for such rigidity results though. The problem is that scal can still tend to zero quite rapidly at infinity. However, if the rate of decay at infinity can be controlled, positive scalar curvature still has profound consequences. For example, Chen has recently shown a decomposition result for 3-manifolds under the assumption that a metric of positive scalar curvature with controlled decay at infinity exists [Chen2025]. This naturally raises the question of how fast the scalar curvature of a given complete Riemannian manifold X decays at infinity. One result in this direction is the following theorem by Gromov [?, p. 15f.].

Theorem (Enlargeable Decay Theorem). *Let X be a complete Riemannian manifold and $V \rightarrow M$ a 2-dimensional real vector bundle over an enlargeable manifold M . Assume that*

- (i) *There exists a proper map $f: X \rightarrow V$, $\deg(f) \neq 0$,*
- (ii) *The total space of the circle bundle corresponding to V is enlargeable.*

Then

$$\min_{B_R(x_0)} \text{scal}(X) \leq \frac{8\pi^2}{(R - R_0)^2}$$

for some $R_0 = R_0(X, x_0)$ and all $R > R_0$, where $B_R(x_0)$ is the ball of radius R around a point $x_0 \in M$.

Condition (ii) is always fulfilled if the bundle $V \rightarrow M$ is trivial (compare Lemma ??), which led Gromov to pose the following question, here written using the notation introduced above.

Question ([?, p. 16]). “What happens to nontrivial bundles $V \rightarrow M$? [...] In general, the examples in [?] indicate a possibility of non-enlargeable circle bundles over enlargeable M ; yet, it seems hard(er) to find such examples, where the corresponding V would admit complete metrics with $\text{scal} \geq \sigma > 0$.”

Note that the existence of such a bundle $V \rightarrow M$ would provide counterexamples to the above decay theorem if condition (ii) is dropped entirely.

As anticipated by Gromov, examples of non-enlargeable circle bundles over enlargeable manifolds do in fact exist. Bundles with this behaviour have recently been constructed in a paper by Kumar and Sen [?] by utilizing tools from symplectic geometry. A very rough sketch of the construction can be given as follows. Take M to be a so-called Donaldson divisor of a suitable closed symplectic manifold, e.g., the torus T^{2n} , $n \geq 3$. Such a Donaldson divisor is a certain codimension 2 symplectic submanifold $M^{2n-2} \subseteq T^{2n}$, which in some cases can be shown to be enlargeable. Now define a circle bundle $E := \partial\nu(M)$ over M as the boundary of a tubular neighbourhood $\nu(M) \subseteq T^{2n}$ of M . The authors then proceed to show that E admits a metric of positive scalar curvature by the following argument. Consider the nullbordism $W := T^{2n} - \nu(M)$ obtained by removing the tubular neighbourhood from the ambient manifold. Now explicitly construct a Morse function $\psi: W \rightarrow \mathbb{R}$ with critical points of index at most n , which shows that E is obtained from the empty set by surgeries of codimension at least $n \geq 3$. Thus, E admits a metric of positive scalar curvature by the Surgery Theorem (see Theorem ??) and so in particular it is not enlargeable (see Theorem ??).

In this thesis, we take a different approach and provide a much more direct construction of such bundles drawing only on general topological concepts, without invoking symplectic methods. This approach, however, is limited to the totally non-spin case. We then apply our results to obtain examples of 2-dimensional vector bundles over enlargeable manifolds, where the total space admits a metric of uniformly positive scalar curvature, thereby providing an affirmative answer to Gromov's question. Note that

The thesis is organized as follows. Chapter 1 serves as a review of principal bundles, classifying spaces and characteristic classes and lays the foundation for the subsequent chapters. In Chapter 2, we briefly introduce the reader to the theory of positive scalar curvature and prove two existence theorems, which are later applied in the main construction. Chapter 3 provides a brief discussion of the concept of enlargeability. With the necessary background established, we then present the main results discussed above in Chapter 4. Finally, Chapter 5 discusses some further questions regarding the preceding construction.

1 Preliminaries

This chapter develops the necessary preliminaries for what follows by recalling the notions of principal bundles, classifying spaces, and characteristic classes. We assume that the reader is familiar with the concept of a fiber bundle with structure group. A classical reference for this is, for example, [?]. Moreover, we presuppose some basic familiarity with homotopy and homology theory. The first section, on principal bundles, draws mainly on [?] and [?]. In the second section, on classifying spaces, we follow [?]. Finally, the third section, on characteristic classes, is based primarily on [?], with additional material drawn from [?] and [?].

1.1 Principal Bundles and Associated Bundles

Definition 1.1. Let G be a topological group. A G -principal bundle is a fiber bundle with fiber G and structure group G acting by left translations.

The total space of a G -principal bundle $P \xrightarrow{\pi} B$ carries a natural right action of G defined as follows. Around $y \in P$, choose a local trivialisation

$$\phi: U \times G \rightarrow \pi^{-1}(U)$$

and let $\phi^{-1}(y) = (\pi(y), h)$. Then set

$$y \cdot g := \phi(\pi(y), hg).$$

This is well defined since, by definition, G acts on itself by *left* translations under a change of local trivialisation. A morphism of principal bundles should respect this right action, thus we define:

Definition 1.2. A morphism from a G -principal bundle $P \rightarrow B$ to an H -principal bundle $P' \rightarrow B'$ is a triple (F, f, ρ) of continuous maps $F: P \rightarrow P'$ and $f: B \rightarrow B'$ together with a homomorphism $\rho: G \rightarrow H$, such that

$$\begin{array}{ccc} P & \xrightarrow{F} & P' \\ \downarrow & & \downarrow \\ B & \xrightarrow{f} & B' \end{array}$$

commutes and such that F is ρ -equivariant, that is $F(x \cdot g) = F(x) \cdot \rho(g)$ for all $x \in P$ and $g \in G$. If $G = H$, we assume ρ to be the identity, and if $B = B'$, we assume f to be the identity.

The condition of equivariance in the above definition is actually quite strong, as illustrated by the following proposition.

Proposition 1.3. *Any morphism of G -principal bundles over the same base space is an isomorphism.*

1 Preliminaries

Proof. First suppose we are given a morphism

$$F: B \times G \rightarrow B \times G$$

of trivial bundles. Then by definition F is of the form

$$F(x, g) = (x, \phi_x(g))$$

for a continuous family of maps $\phi_x: G \rightarrow G$. By equivariance, $\phi_x(g) = \phi_x(e)g$, and thus an inverse of F is given by

$$F^{-1}(x, g) := (x, \phi_x(e)^{-1}g).$$

Since every G -principal bundle is locally trivial, this shows that we can always find local inverses in the general case. As inverses are unique, they fit together to define a global inverse. $\square$

To any fiber bundle with structure group G , one can canonically associate a G -principal bundle by a general construction that allows one to change the fiber of a given bundle. We now recall this construction of changing the fiber. Let $p: E \rightarrow B$ be a fiber bundle with fiber F and structure group G and let F' be another space upon which G acts effectively. Then we can construct a new bundle as follows. Choose an open cover $(U_\alpha)_{\alpha \in A}$ of B together with local trivialisations $\phi_\alpha: U_\alpha \times F \rightarrow p^{-1}(U_\alpha)$. Let $\tau_{\alpha\beta}: U_\alpha \cap U_\beta \rightarrow G$ be the corresponding transition functions defined by

$$(\phi_\alpha^{-1} \circ \phi_\beta)(x, f) = (x, \tau_{\alpha\beta}(x) \cdot f).$$

Define

$$E' := \left(\coprod_{\alpha \in A} U_\alpha \times F' \right) / \sim$$

where the equivalence relation $\sim$ is defined by

$$(b, f', \beta) \sim (b, \tau_{\alpha\beta}(b) \cdot f', \alpha)$$

and let $p': E' \rightarrow B$, $[(b, f', \alpha)] \mapsto b$. Then the maps

$$\psi_\alpha: U_\alpha \times F' \rightarrow (p')^{-1}(U_\alpha), (b, f') \mapsto [(b, f', \alpha)].$$

define local trivialisations of p' with the same transition functions $\tau_{\alpha\beta}$, giving $p': E' \rightarrow B$ the structure of a fiber bundle with fiber F' and structure group G . We say that this new bundle is obtained from $E \rightarrow B$ by *changing the fiber* from F to F' (with the G -action on F' being implicit). Note that we can also obtain $E \rightarrow B$ from $E' \rightarrow B$ by changing the fiber from F' back to F . Thus no information is lost in this process. Considering the special case $F' = G$ with G acting on itself by left translations, we call the bundle obtained from $E \rightarrow B$ by changing the fiber from F to G the *associated principal bundle*. Morally speaking, if we want to understand the bundle $E \rightarrow B$, it is sufficient to understand its associated principal bundle, since this contains the same information, as we have seen above. The benefit of studying principal bundles is that they are highly symmetric objects, and thus usually easier to understand. For example, many constructions simplify for principal bundles, such as the concept of universal bundles, as we will see in the next section. Another such instance is the construction for changing the fiber from above. In the case of principal bundles this can be carried out more directly and has the benefit of also allowing for non-effective group actions. This is known as the *Borel Construction*.

Lemma 1.4. *Let $p: P \rightarrow B$ be a G -principal bundle and F a space upon which G acts from the left. Then*

$$E := (P \times F)/\sim, \quad (y, f) \sim (y \cdot g, g^{-1}f)$$

together with the map $\pi: E \rightarrow B$ given by $[(y, f)] \mapsto p(y)$ is a fiber bundle with fiber F and structure group G . Moreover, if the action of G on F is effective, this bundle is isomorphic to the bundle obtained from $P \rightarrow B$ by changing the fiber from G to F using the construction outlined above.

Proof. To see that $P \times_G F$ is locally trivial, let $\psi: U \times G \rightarrow P$ be a local trivialisation. Then the homeomorphism

$$\psi \times \text{id}_F: U \times G \times F \rightarrow p^{-1}(U) \times F$$

descends to a homeomorphism

$$\tilde{\kappa}: (U \times G \times F)/\sim \rightarrow (p^{-1}(U) \times F)/\sim = \pi^{-1}(U),$$

where the equivalence relation on $U \times G \times F$ is given by $(x, g, f) \sim (x, gh, h^{-1}f)$. Moreover, the map

$$U \times G \times F \rightarrow U \times F, \quad (x, g, f) \mapsto (x, gf)$$

descends to a homeomorphism $(U \times G \times F)/\sim \rightarrow U \times F$ with inverse given by $(x, f) \mapsto [(x, e, f)]$. Under this homeomorphism, $\tilde{\kappa}$ corresponds to a homeomorphism $\kappa: U \times F \rightarrow \pi^{-1}(U)$, which is explicitly given by $(x, f) \mapsto [(\psi(x, e), f)]$. Since

$$\begin{array}{ccc} U \times F & \xrightarrow{\kappa} & \pi^{-1}(U) \\ & \searrow \text{proj.} & \downarrow \pi \\ & & U \end{array}$$

commutes, this shows that $\pi: E \rightarrow B$ is locally trivial. To see that, for effective G -actions, this construction agrees with the change-of-fiber construction introduced above, we compute the transition functions of the local trivialisations just defined. Let $U_\alpha, U_\beta \subseteq B$ be two trivialising neighbourhoods for $P \rightarrow B$ with local trivialisations ψ_α, ψ_β and transition function $\tau_{\alpha\beta}: U_\alpha \cap U_\beta \rightarrow G$. Moreover, let $\kappa_\alpha, \kappa_\beta$ be the corresponding local trivialisations for $E \rightarrow B$ as constructed above. Then we compute

$$(\kappa_\alpha^{-1} \circ \kappa_\beta)(x, f) = \kappa_\alpha^{-1}([\psi_\beta(x, e), f]) = (x, \tau_{\alpha\beta}(x)f).$$

Thus the statement follows from the fact that fiber bundles, where the structure group acts effectively, are uniquely determined by the transition functions of local trivialisations. $\square$

We denote the bundle $E \rightarrow B$ obtained from the above construction by $P \times_G F \rightarrow B$. The Borel construction is natural with respect to pullbacks, as stated in the following lemma.

Lemma 1.5. *Let $p: P \rightarrow B$ be a G -principal bundle, F a space upon which G acts from the left and $f: B' \rightarrow B$ a continuous map. Then*

$$(f^*P) \times_G F \cong f^*(P \times_G F).$$

Proof. An explicit model for the pullback bundle is

$$f^*P := \{(y, b') \in P \times B' \mid p(y) = f(b')\}.$$

Then

$$(f^*P) \times_G F = \{(y, b', f) \in P \times B' \times F \mid p(y) = f(b')\} / \sim,$$

for $(y, b', f) \sim (y \cdot g, b', g^{-1} \cdot f)$. Moreover,

$$\begin{aligned} f^*(P \times_G F) &= \{([y, f], b') \in P \times_G F \times B' \mid p(y) = f(b')\} \\ &\cong \{(y, f, b') \in P \times F \times B' \mid p(y) = f(b')\} / \sim \end{aligned}$$

for $(y, f, b') \sim (y \cdot g, g^{-1} \cdot f, b')$. Clearly these are isomorphic bundles. $\square$

Remark 1.6. We remark that the action of G on F in the above lemma need not be effective. If we restrict ourselves to effective actions, a more general naturality result holds for the change-of-fiber construction of arbitrary bundles. Namely, first pulling a bundle back under some continuous map and then changing the fiber amounts to the same as first changing the fiber and then pulling back. Comparing the transition functions of the local trivialisations in both cases shows that they in fact agree.

Let's look at some concrete examples for the theory developed above.

Example 1.7. (i) Let $\pi: E \rightarrow B$ be a real rank n vector bundle over a paracompact base space B . Choosing a bundle metric gives $E \rightarrow B$ the structure of a fiber bundle with fiber $\mathbb{R}^n$ and structure group $O(n)$. The associated principal bundle is explicitly given by the *orthonormal frame bundle* $O(E)$ of E , whose fiber over a point $x \in B$ is the space of all ordered orthonormal bases of E_x . The topology on $O(E)$ is the subspace topology $O(E) \subseteq E \times \dots \times E$. Local trivialisations of the canonical projection map $p: O(E) \rightarrow B$ can be defined as follows. Choose an orthonormal local frame $e_1, \dots, e_n$ of E defined over an open set $U \subseteq B$. Then a trivialisation $\phi: U \times O(n) \rightarrow p^{-1}(U)$ is given by

$$(x, A) \mapsto \left(\sum_{i=1}^n a_{i1} e_i(x), \dots, \sum_{i=1}^n a_{in} e_i(x) \right),$$

where $A = (a_{ij})$. So $\phi(x, A)$ is the basis of $E_x = \pi^{-1}(x)$, whose coordinates in the frame (e_i) are precisely the columns of the matrix A . The transition functions of these trivialisations agree with the transition functions of the trivialisations of E . Since the transition functions uniquely determine the bundle, we conclude that $O(E)$ is in fact isomorphic to the associated principal bundle as defined at the beginning of this section. Analogously, if $E \rightarrow B$ is a complex rank n vector bundle, choosing a hermitian metric on E allows us to reduce the structure group to $U(n)$, and the corresponding associated principal bundle is the *unitary frame bundle*.

(ii) Let $p: \tau_1^n \rightarrow \mathbb{C}P^n$ be the tautological complex line bundle equipped with the canonical hermitian metric. Since this bundle is of rank 1, its unitary frame bundle is simply the subspace $U(\tau_1^n) \subseteq \tau_1^n$ of all vectors of norm 1, together with the restriction of p to this subspace. Recall that

$$\tau_1^n = \{(L, z) \in \mathbb{C}P^n \times \mathbb{C}^{n+1} \mid z \in L\}.$$

So, explicitly,

$$U(\tau_1^n) = \{(L, z) \in \tau_1^n \mid |z| = 1\}.$$

Consider the diffeomorphism

$$U(\tau_1^n) \rightarrow S^{2n+1}, (L, z) \mapsto z$$

with inverse $S^{2n+1} \rightarrow U(\tau_1^n)$, $z \mapsto (\pi(z), z)$, where $\pi: S^{2n+1} \rightarrow \mathbb{C}P^n$ is the canonical projection. Under this diffeomorphism the restriction of p to $U(\tau_1^n)$ corresponds to π . Hence π defines a $U(1) \cong S^1$ -principal bundle, which agrees with the associated principal bundle of $\tau_1^n \rightarrow \mathbb{C}P^n$.

- (iii) Let $E \rightarrow B$ be an S^1 -principal bundle over a smooth manifold B without boundary. We can change the fiber of E from S^1 to D^2 to obtain another bundle $DE \rightarrow B$, where S^1 acts on D^2 by rotations. By definition of the change-of-fiber construction, trivialisations $U \times D^2 \rightarrow DE$ have the same transition functions as the corresponding trivialisations of $E \rightarrow B$. Moreover, they restrict to trivialisations $U \times S^1 \rightarrow \partial DE$ of the boundary. Since the action of S^1 on D^2 restricts to the action of S^1 on itself, we see that $E \cong \partial DE$. So, in particular, the total space of any S^1 -principal bundle is null-bordant. We will make use of this fact in Chapter 4.

1.2 Universal Bundles and Classifying Spaces

A natural question is which fiber bundles with specified fiber and structure group can exist over a given base space. As before, this problem turns out to be easier for principal bundles. We assume all base spaces to be CW-complexes in this section.

Definition 1.8. A G -principal bundle $E \rightarrow B$ is called *universal* if E is weakly contractible as a topological space.

The significance of this definition is established by the following theorem. We say that two morphisms of G -principal bundles are G -homotopic if they are homotopic through morphisms of G -principal bundles.

Theorem 1.9. *Let $E \rightarrow B$ be a universal G -principal bundle. Then for any other G -principal bundle $P \rightarrow X$ there exists a morphism $P \rightarrow E$ of G -principal bundles, which is unique up to G -homotopy.*

Before showing the theorem, let us discuss its implications. Firstly, we can equivalently formulate the theorem by saying that $E \rightarrow B$ is a terminal object in the homotopy category of G -principal bundles over CW-complexes and thus, in particular, uniquely determined (up to G -homotopy equivalence). In view of this, one usually writes $EG := E$ and $BG := B$, and refers to $EG \rightarrow BG$ as *the* universal G -principal bundle. The base space BG is called the *classifying space* of G . Since any bundle morphism $(F, f): P \rightarrow EG$ induces a morphism $P \rightarrow f^*EG$ (and vice versa, which is true for arbitrary fiber bundles), we conclude from Proposition 1.3 that

$$P \cong f^*EG.$$

We call f a *classifying map* for P . By homotopy invariance of the pullback, we get a well-defined bijection

$$\phi_X: [X, BG] \rightarrow \mathcal{P}_G(X), [f] \mapsto [f^*EG],$$

where $\mathcal{P}_G(X)$ denotes the set of isomorphism classes of G -principal bundles over X . The existence result in Theorem 1.9 shows that ϕ_X is surjective and the uniqueness result shows that it is injective. Moreover, ϕ_X is clearly natural in X . Thus we have shown:

Corollary 1.10. *The maps ϕ_X define a natural isomorphism of contravariant functors*

$$[-, BG], \mathcal{P}_G(-): \mathbf{HCW} \rightarrow \mathbf{Set},$$

where $\mathbf{HCW}$ denotes the homotopy category of CW-complexes.

Theorem 1.9 will be a simple application of the following theorem.

Theorem 1.11. *Let $P \rightarrow X$ be a G -principal bundle and $A \subseteq X$ a subcomplex. Then any morphism $P|_A \rightarrow EG$ of G -principal bundles extends to a morphism $P \rightarrow EG$ of G -principal bundles.*

1 Preliminaries

The proof of this theorem is based on an inductive, cell-by-cell approach. We refer the reader to [?, Theorem 2.7.1].

Proof of Theorem 1.9. Applying Theorem 1.11 to $A = \emptyset$ shows the existence of a morphism $P \rightarrow EG$. Now suppose we are given morphisms $F_0, F_1: P \rightarrow EG$. Consider the G -principal bundle

$$\pi \times \text{id}: P \times [0, 1] \rightarrow X \times [0, 1],$$

where $\pi: P \rightarrow X$ is the bundle projection (this is just the pullback bundle of P under the projection $X \times [0, 1] \rightarrow X$). Then F_0 and F_1 define a morphism of G -principal bundles

$$(F_0, F_1): P \times \{0, 1\} = (P \times [0, 1])|_{X \times \{0, 1\}} \rightarrow EG,$$

which by Theorem 1.11 can be extended to $P \times [0, 1]$, thus yielding a G -homotopy between F_0 and F_1 . $\square$

Of course, for this theory to be of any use, we need such universal bundles to exist. The following general existence theorem was first shown by Milnor [?].

Theorem 1.12. *For any topological group G there exists a universal G -principal bundle $EG \rightarrow BG$, where BG is a CW-complex.*

A priori, the space BG constructed in [?] may not be a CW-complex. However, if we choose a CW-approximation $f: X \rightarrow BG$, the total space of the bundle $f^*EG \rightarrow X$ is still weakly contractible. This follows by applying the five lemma to the following diagram.

$$\begin{array}{ccccccccc} \pi_{k+1}(X) & \longrightarrow & \pi_k(G) & \longrightarrow & \pi_k(f^*EG) & \longrightarrow & \pi_k(X) & \longrightarrow & \pi_{k-1}(G) \\ \downarrow \cong & & \downarrow \text{id} & & \downarrow & & \downarrow \cong & & \downarrow \text{id} \\ \pi_{k+1}(BG) & \longrightarrow & \pi_k(G) & \longrightarrow & \pi_k(EG) & \longrightarrow & \pi_k(BG) & \longrightarrow & \pi_{k-1}(G) \end{array}$$

We can now use these classification results for principal bundles to obtain analogous results for general fiber bundles by appealing to the change-of-fiber construction discussed in the preceding section.

Corollary 1.13. *Let G be a topological group acting effectively on a space F . Then for any CW-complex X the map*

$$[X, BG] \rightarrow \mathcal{F}_{F,G}(X), \quad [f] \mapsto [f^*(EG \times_G F)]$$

is a natural bijection, where $\mathcal{F}_{F,G}(X)$ denotes the set of isomorphism classes of fiber bundles with fiber F and structure group G over X .

Proof. By Corollary 1.10, the map

$$[X, BG] \rightarrow \mathcal{P}_G(X), \quad [f] \mapsto [f^*EG]$$

is a natural bijection, and by Lemma 1.5

$$\mathcal{P}_G(X) \rightarrow \mathcal{F}_{F,G}(X), \quad [P] \mapsto [P \times_G F]$$

is a natural bijection. Composing both thus yields the natural bijection

$$[X, BG] \rightarrow \mathcal{F}_{F,G}(X), \quad [f] \mapsto [(f^*EG) \times_G F].$$

So the statement follows by again applying Lemma 1.5. $\square$

Again let us discuss a few examples.

Example 1.14.

- (i) The universal cover $\mathbb{R} \rightarrow S^1$ is a $\mathbb{Z}$ -principal bundle, $\mathbb{Z}$ acting on itself by addition. Since $\mathbb{R}$ is contractible we obtain $B\mathbb{Z} \cong S^1$ and $E\mathbb{Z} \cong \mathbb{R}$.
- (ii) From the long exact sequence of homotopy groups

$$\pi_k(EG) \longrightarrow \pi_k(BG) \longrightarrow \pi_{k-1}(G) \longrightarrow \pi_{k-1}(EG)$$

we conclude that $\pi_k(BG) \cong \pi_{k-1}(G)$ for $k \geq 1$. Moreover, BG is connected since it is the image of the connected space EG under $EG \rightarrow BG$.

- (iii) The universal $O(k)$ -principal bundle $EO(k) \rightarrow BO(k)$ also has an explicit description. Consider the Stiefel manifold $V_k^n \subseteq \mathbb{R}^{n \times k}$, which is defined as the compact space of orthonormal k -tuples of vectors in $\mathbb{R}^n$. Moreover let Gr_k^n be the Grassmannian of k -planes in $\mathbb{R}^n$, i.e. the set of k -dimensional linear subspaces of $\mathbb{R}^n$. The topology on Gr_k^n is induced by the projection map

$$p: V_k^n \rightarrow \text{Gr}_k^n, (v_1, \dots, v_k) \mapsto \text{span}(v_1, \dots, v_k)$$

Identifying a linear subspace $V \in \text{Gr}_k^n$ with $\mathbb{R}^k$ yields an identification of the fiber $p^{-1}(V)$ with $O(k)$. Moreover the identifications $V \cong \mathbb{R}^k$ can be chosen consistently in a small neighbourhood around V , providing local trivialisations for p . The transition of two such trivialisations is realized by actions of $O(k)$, hence p defines an $O(k)$ -principal bundle. Now the canonical inclusion $\mathbb{R}^n \rightarrow \mathbb{R}^{n+1}$ induces inclusions $V_k^n \rightarrow V_k^{n+1}$ and $\text{Gr}_k^n \rightarrow \text{Gr}_k^{n+1}$. Consider the colimits over these inclusions

$$V_k^\infty := \text{colim}_n (V_k^n), \quad \text{Gr}_k^\infty := \text{colim}_n (\text{Gr}_k^n)$$

and the induced map $V_k^\infty \rightarrow \text{Gr}_k^\infty$. One can choose trivialisations of $V_k^n \rightarrow \text{Gr}_k^n$ for varying n , which fit together to induce a trivialisation in the colimit, such that $V_k^\infty \rightarrow \text{Gr}_k^\infty$ defines an $O(k)$ -principal bundle. We claim that V_k^∞ is weakly contractible. Consider the map

$$V_k^n \rightarrow S^{n-1}, (v_1, \dots, v_k) \mapsto v_1.$$

Similar arguments as above shows that this map is locally trivial with fiber V_{k-1}^{n-1} . Hence we can consider the long exact sequence of homotopy groups

$$\pi_{i+1}(S^{n-1}) \longrightarrow \pi_i(V_{k-1}^{n-1}) \longrightarrow \pi_i(V_k^n) \longrightarrow \pi_i(S^{n-1})$$

So $\pi_i(V_k^n) \cong \pi_i(V_{k-1}^{n-1})$ for $i \leq n-3$. Applying this inductively yields

$$\pi_i(V_k^n) \cong \pi_i(V_1^{n-(k-1)}) = \pi_i(S^{n-k+1}) = 0$$

for $i \leq n-k-1$. Now from compactness of S^i and the fact that V_k^n is closed in V_k^{n+1} we conclude that the image of any map $S^i \rightarrow V_k^\infty$ is contained in some finite stage V_k^n . Therefore $\pi_i(V_k^\infty) = 0$ for all $i \geq 0$, such that the universal $O(k)$ -principal bundle is explicitly given by $V_k^\infty \rightarrow \text{Gr}_k^\infty$.

Similarly we get an explicit description of the universal $SO(k)$ -principal bundle. Let $\tilde{\text{Gr}}_k^n$ be the set of tuples consisting of a k -dimensional linear subspace of $\mathbb{R}^n$ together with a chosen orientation. The topology on $\tilde{\text{Gr}}_k^n$ is induced by

$$p: V_k^n \rightarrow \tilde{\text{Gr}}_k^n, (v_1, \dots, v_k) \mapsto (\text{span}(v_1, \dots, v_k), o_{(v_1, \dots, v_k)}),$$

where $o_{(v_1, \dots, v_k)}$ is the unique orientation on $\text{span}(v_1, \dots, v_k)$, such that $(v_1, \dots, v_k)$ is positively oriented. Similar to above we see that p defines an $SO(k)$ -principal bundle. Taking the colimit, we obtain the universal $SO(k)$ -principal bundle $V_k^\infty \rightarrow \tilde{\text{Gr}}_k^\infty$.

- (iv) Any real k -dimensional vector bundle $E \rightarrow X$ can be considered as a fiber bundle with fiber $\mathbb{R}^k$ and structure group $O(k)$. So Corollary 1.13 and the example above show that every such E is the pullback of the universal real k -dimensional vector bundle $V_k^\infty \times_{O(k)} \mathbb{R}^k$ under some map $X \rightarrow \text{Gr}_k^\infty$. We show that this universal vector bundle also has a more explicit description, namely as the tautological bundle $\gamma_k \rightarrow \text{Gr}_k^\infty$. Recall that this bundle is defined as

$$\gamma_k := \{(X, v) \in \text{Gr}_k^\infty \times \mathbb{R}^\infty \mid v \in X\}$$

together with the canonical projection map to Gr_k^∞ . An element in V_k^∞ is a k -tuple of orthonormal vectors in some $\mathbb{R}^n$, which we will subsequently regard as the columns of an $n \times k$ matrix. Then the map

$$V_k^\infty \times \mathbb{R}^k \rightarrow \gamma_k, (A, v) \mapsto (\text{im}(A), Av)$$

descends to the quotient, defining a vector bundle map $V_k^\infty \times_{O(k)} \mathbb{R}^k \rightarrow \gamma_k$. This map is clearly a linear isomorphism on fibers, hence an isomorphism of vector bundles.

Similiarly, oriented real k -dimensional vector bundles are classified by maps to $\tilde{\text{Gr}}_k^\infty$. We claim that the universal oriented real k -dimensional vector bundle is given by the tautological bundle $\gamma_k^{SO} \rightarrow \tilde{\text{Gr}}_k^\infty$ defined by

$$\gamma_k^{SO} := \{(X, o_X, v) \in \tilde{\text{Gr}}_k^\infty \times \mathbb{R}^k \mid v \in X\}$$

together with the canonical projection map to $\tilde{\text{Gr}}_k^\infty$. To this end consider the map

$$V_k^\infty \times \mathbb{R}^k \rightarrow \gamma_k^{SO}, (A, v) \mapsto (\text{im}(A), o_A, Av),$$

where o_A is the unique orientation on $\text{im}(A)$, such that $A: \mathbb{R}^k \rightarrow \text{im}(A)$ is orientation preserving. Again this map descends to the quotient defining an isomorphism $V_k^\infty \times_{SO(k)} \mathbb{R}^k \rightarrow \gamma_k^{SO}$.

- (v) Completely analogous to the real case we can also define the complex Stiefel manifolds $V_k^\infty(\mathbb{C})$ and complex Grassmanians $\text{Gr}_k^\infty(\mathbb{C})$ and show that the universal $U(k)$ -principal bundle is given by $V_k^\infty(\mathbb{C}) \rightarrow \text{Gr}_k^\infty(\mathbb{C})$. The universal complex k -dimensional vector bundle is the tautological bundle $\tau_k \rightarrow \text{Gr}_k^\infty(\mathbb{C})$.
- (vi) As special cases of the preceeding examples we obtain

$$B\mathbb{Z}_2 \cong \mathbb{R}P^\infty, \quad BS^1 \cong \mathbb{C}P^\infty.$$

1.3 Stiefel-Whitney and Chern-Classes

In the preceding chapter we discussed the question of how bundles can be classified. Another fundamental question is, of course, how to determine whether two given bundles are isomorphic. As usual in topology, a partial solution to this problem is provided by defining computable invariants. For vector bundles the most important invariants are so-called *characteristic classes*. The general definition of a characteristic class is as follows. Let $\mathbb{K} = \mathbb{R}, \mathbb{C}$ and write $\text{Vect}_{\mathbb{K}}(B)$ for the set of isomorphism classes of $\mathbb{K}$ -vector bundles over a space B . Then $\text{Vect}_{\mathbb{K}}(-)$ defines a contravariant functor $\text{Top} \rightarrow \text{Set}$. A characteristic class c for $\mathbb{K}$ -vector bundles with values in a cohomology theory H^* is a natural transformation from the functor $\text{Vect}_{\mathbb{K}}(-)$ to the functor H^* , regarded as a functor with values in Set . In more concrete terms, c assigns to each $\mathbb{K}$ -vector bundle $E \rightarrow B$ a cohomology class $c(E) \in H^*(B)$, depending only on the isomorphism class of E , such that for any continuous map $f: B' \rightarrow B$ the following naturality condition holds:

$$c(f^*E) = f^*c(E).$$

Using the theory of classifying spaces developed above, it follows that for real (resp. complex) k -dimensional vector bundles, the characteristic class c is uniquely determined by its value on the universal bundle γ_k (resp. τ_k) discussed in Example 1.14 (iv) (resp. ??).

For real vector bundles the most important characteristic classes are the *Stiefel-Whitney classes* and for complex vector bundles the *Chern classes*. Both are characterized by a few axioms.

Definition 1.15 (Stiefel-Whitney Classes, HATCHER). There is a unique sequence of characteristic classes $w_1, w_2, \dots$ for real vector bundles with values in $H^*(-; \mathbb{Z}_2)$, such that

- (i) $w_i(E) \in H^i(B; \mathbb{Z}_2)$ for any real bundle $E \rightarrow B$,
- (ii) $w_i(E) = 0$ for $i > \text{rank}(E)$,
- (iii) $w(E_1 \oplus E_2) = w(E_1) \smile w(E_2)$, where $w = 1 + w_1 + w_2 + \dots$,
- (iv) For the universal real line bundle $\gamma_1 \rightarrow \mathbb{R}P^\infty$, $w_1(\gamma_1)$ is the generator of $H^1(\mathbb{R}P^\infty; \mathbb{Z}_2) \cong \mathbb{Z}_2$.

Note that by property (ii) the sum $w = 1 + w_1 + w_2 + \dots$ is always finite. The class $w(E)$ is called the *total Stiefel-Whitney class* of E . We sometimes write $w_0(E) := 1 \in H^0(B; \mathbb{Z}_2)$ to simplify notation.

Definition 1.16 (Chern Classes, HATCHER). There is a unique sequence of characteristic classes $c_1, c_2, \dots$ for complex vector bundles with values in $H^*(-; \mathbb{Z})$, such that

- (i) $c_i(E) \in H^{2i}(B; \mathbb{Z})$ for any complex bundle $E \rightarrow B$,
- (ii) $c_i(E) = 0$ for $i > \text{rank}(E)$,
- (iii) $c(E_1 \oplus E_2) = c(E_1) \smile c(E_2)$, where $c = 1 + c_1 + c_2 + \dots$,
- (iv) For the universal complex line bundle $\tau_1 \rightarrow \mathbb{C}P^\infty$, $c_1(\tau_1)$ is a generator of $H^2(\mathbb{C}P^\infty; \mathbb{Z}) \cong \mathbb{Z}$ specified in advance.

Again the *total Chern class* $c = 1 + c_1 + c_2 + \dots$ is well-defined and we write $c_0(E) := 1 \in H^0(B; \mathbb{Z})$. Note that in both definitions, property (iv) uniquely determines the values of the respective characteristic classes on line bundles and, together with property (iii), prescribes their values on bundles that split as sums of line bundles. The values on arbitrary bundles can then be deduced from a general splitting principle [??]. We refer the reader to [HATCHER] for a detailed proof of existence and uniqueness.

Not only are the definitions of Stiefel-Whitney and Chern classes formally quite similar, but there is also a direct connection between the two.

Proposition 1.17 (Hatcher Proposition 3.8). *Regarding a complex rank n vector bundle $E \rightarrow B$ as a real rank $2n$ vector bundle, $w_{2i+1}(E) = 0$ and $w_{2i}(E)$ is the image of $c_i(E)$ under the coefficient homomorphism $H^{2i}(B; \mathbb{Z}) \rightarrow H^{2i}(B; \mathbb{Z}_2)$.*

Let us consider some examples.

Example 1.18.

- (i) Let $\iota: \mathbb{R}P^n \rightarrow \mathbb{R}P^\infty$ be the canonical inclusion. Then the pullback $\iota^*\gamma_1$ is isomorphic to the tautological line bundle $\gamma_1^n \rightarrow \mathbb{R}P^n$. Since ι induces an isomorphism on cohomology we conclude from naturality that $w_1(\gamma_1^n) = \iota^*w_1(\gamma_1)$ is the generator of $H^1(\mathbb{R}P^n; \mathbb{Z}_2) \cong \mathbb{Z}_2$. Similarly, $c_1(\tau_1^n)$ of the tautological complex line bundle $\tau_1^n \rightarrow \mathbb{C}P^n$ is a generator of $H^2(\mathbb{C}P^n; \mathbb{Z}) \cong \mathbb{Z}$.

1 Preliminaries

- (ii) Let $\varepsilon \rightarrow B$ be a trivial real bundle over some space B . Then $\varepsilon \cong f^*\varepsilon$ for a constant map $f: B \rightarrow B$ and thus $w_i(\varepsilon_n) = f^*w_i(\varepsilon) = 0$ for all $i \geq 1$. Similarly all Chern classes for complex trivial bundles vanish.
- (iii) Let $TS^n \rightarrow S^n$ be the tangent bundle of the sphere and let $\nu_n \rightarrow S^n$ be the normal bundle with respect to the standard embedding $S^n \rightarrow \mathbb{R}^{n+1}$. Since ν_n and $TS^n \oplus \nu_n$ are trivial we conclude from the example above and property (iii) in Definition 1.15 that

$$1 = w(TS^n \oplus \nu_n) = w(TS^n) \smile w(\nu_n) = w(TS^n) \smile 1 = w(TS^n).$$

Thus $w_i(TS^n) = 0$ for all $i \geq 1$.

In the last example we saw that although all Stiefel-Whitney classes of a bundle vanish, it need not be trivial. However for one-dimensional real (resp. complex) bundles it turns out that the first Stiefel-Whitney (resp. Chern) class already completely determines the bundle:

Theorem 1.19 (Hatcher, Proposition 3.10). *Let X be a CW-complex. Then the maps*

$$w_1: \text{Vect}_{\mathbb{R}}(X) \rightarrow H^1(X; \mathbb{Z}_2)$$

and

$$c_1: \text{Vect}_{\mathbb{C}}(X) \rightarrow H^2(X; \mathbb{Z})$$

are bijections.

While the first Stiefel-Whitney class of higher dimensional bundles does not identify their isomorphism type it does characterize another important invariant, namely orientability:

Proposition 1.20 (Hatcher, Prop 3.11). *Let X be a CW-complex. A real vector bundle $E \rightarrow X$ is orientable if and only if $w_1(E) = 0$.*

Similarly vanishing of the second Stiefel Whitney class corresponds to the existence of another structure, namely that of a *spin structure*.

Definition 1.21. (i) For $n \geq 2$ the *spin group* $\text{Spin}(n)$ is defined as the total space of the non-trivial two-sheeted cover $\rho: \text{Spin}(n) \rightarrow \text{SO}(n)$. If $n \geq 3$ this is the universal cover and in the case $n = 2$ we have $\text{Spin}(2) \cong \text{SO}(2)$. For $n = 1$ we define $\text{Spin}(1) := \mathbb{Z}_2$, the total space of the two-sheeted cover $\rho: \mathbb{Z}_2 \rightarrow \text{SO}(1)$.

- (ii) Let $E \rightarrow B$ be an oriented real vector bundle equipped with a fiber metric and let $P_{\text{SO}(n)}$ be its oriented orthonormal frame bundle. A *spin structure* on E is a $\text{Spin}(n)$ -principal bundle $P_{\text{Spin}(n)}$ together with a map

$$\phi: P_{\text{Spin}(n)} \rightarrow P_{\text{SO}(n)}$$

of principal bundles, which is ρ -equivariant (compare Definition 1.2).

- (iii) An orientable manifold M is called *spin* if its tangent bundle TM admits a spin structure and is called *totally non-spin* if the universal cover of M is not spin.

Remark 1.22. When $n = 1$, $P_{\text{SO}(1)} \cong X$ and a spin structure is simply a two-sheeted cover of X .

Note that the existence of a spin structure does not depend on the chosen orientation or fiber metric on E , since different choices yield isomorphic oriented orthonormal frame bundles. In fact the only obstruction to the existence of a spin structure is the second Stiefel-Whitney class, as mentioned above.

Theorem 1.23 (Lawson, Micheaslon). *Let X be a CW-complex. An orientable real vector bundle $E \rightarrow X$ admits a spin structure if and only if $w_2(E) = 0$.*

Example 1.24. (i) All spheres S^n , $n \geq 1$ are spin by Example 1.18 (iii).

(ii) The standard example for non-spin manifolds are all even dimensional complex projective spaces $\mathbb{C}P^{2k}$, see [MilnorStasheff??].

Remark 1.25. Let $E \rightarrow X$ be a real vector bundle over a CW-complex. If E is orientable, we have seen that this bundle is classified by a map $X \rightarrow BSO(n)$. Similarly, if a spin structure exists, E is classified by a map $X \rightarrow BSpin(n)$, which can be seen as follows. Let $\phi: P_{Spin(n)} \rightarrow P_{SO(n)}$ be a spin structure on E and let $f: X \rightarrow BSpin(n)$ be a classifying map for $P_{Spin(n)}$. Define a vector bundle

$$\gamma_n^{Spin} := ESpin(n) \times_{Spin(n)} \mathbb{R}^n$$

using Lemma 1.4, where $Spin(n)$ acts on $\mathbb{R}^n$ by virtue of the covering map $\rho: Spin(n) \rightarrow SO(n)$ and the canonical action of $SO(n)$. From Lemma 1.5 we obtain

$$f^* \gamma_n^{Spin} \cong P_{Spin(n)} \times_{Spin(n)} \mathbb{R}^n.$$

Moreover, the spin structure ϕ induces an isomorphism $P_{Spin(n)} \times_{Spin(n)} \mathbb{R}^n \rightarrow P_{SO(n)} \times_{SO(n)} \mathbb{R}^n \cong E$ of vector bundles by

$$[(x, v)] \mapsto [(\phi(x), v)].$$

This map is clearly a linear bijection on fibers, hence an isomorphism of vector bundles.

Recall that a manifold M is orientable if and only if its tangent bundle TM restricted to any embedded loop $S^1 \subseteq M$ is trivial. Or in other words “ TM does not contain any Moebius Bands”. A similar characterization can be given for spin manifolds, which is why the property of being spin is sometimes referred to as a higher dimensional analogue of orientability.

Proposition 1.26 (Georg Paper, Proposition 2.6). *Let M be a connected oriented manifold of dimension at least 5.*

- (i) *M is spin if and only if for every embedded surface $S \subseteq M$ the restriction $TM|_S$ of the tangent bundle to S is trivial.*
- (ii) *M is totally non-spin if and only if there exists an embedding $\iota: S^2 \rightarrow M$, such that $w_2(\iota^*TM) \neq 0$. In particular, the restriction ι^*TM of the tangent bundle to this embedded sphere is non-trivial.*

Proof. Since we will not be making use of the first characterization, we shall only proof part (ii) of the proposition and refer the reader to [Georg Paper] for a proof of part (i). Let $\tilde{M}$ be the universal cover of M . Since $T\tilde{M}$ does not admit a spin structure we conclude from Theorem 1.23 that $w_2(T\tilde{M}) \neq 0$. So by the universal coefficient theorem there exists a homology class $a \in H_2(\tilde{M}; \mathbb{Z}_2)$ with

$$\langle w_2(T\tilde{M}), a \rangle \neq 0.$$

Once more applying the universal coefficient theorem as well as the Hurewicz theorem we obtain an isomorphism

$$\pi_2(\tilde{M}) \otimes \mathbb{Z}_2 \rightarrow H_2(\tilde{M}; \mathbb{Z}) \otimes \mathbb{Z}_2 \cong H_2(\tilde{M}; \mathbb{Z}_2).$$

Explicitly, for some class $\alpha \in \pi_2(\tilde{M})$ represented by a map $f: S^2 \rightarrow \tilde{M}$ the above isomorphism maps $\alpha \otimes 1$ to $f_*[S^2]_{\mathbb{Z}_2}$, where $[S^2]_{\mathbb{Z}_2}$ is the $\mathbb{Z}_2$ -fundamental class of S^2 . Therefore

1 Preliminaries

we can write $a = f_*[S^2]_{\mathbb{Z}_2}$ for some $f: S^2 \rightarrow \tilde{M}$. Let $p: \tilde{M} \rightarrow M$ be the projection and define $\iota := p \circ f: S^2 \rightarrow M$. Since $\dim(M) \geq 5$ we may assume ι to be an embedding. We compute

$$\langle w_2(\iota^*TM), [S^2]_{\mathbb{Z}_2} \rangle = \langle f^*w_2(p^*TM), [S^2]_{\mathbb{Z}_2} \rangle = \langle w_2(T\tilde{M}), f_*[S^2]_{\mathbb{Z}_2} \rangle = \langle w_2(T\tilde{M}), a \rangle \neq 0,$$

where we used $p^*TM \cong T\tilde{M}$ and naturality of the Kronecker pairing in the second equation. Therefore $w_2(\iota^*TM) \neq 0$. $\square$

Finally we record another scenario in which the vanishing of all Stiefel-Whitney classes of a bundle is sufficient for triviality.

Lemma 1.27. *Let $E \rightarrow S$ be a real vector bundle of rank at least 3 over a two dimensional CW-complex S . Then E is trivial if and only if $w_1(E) = w_2(E) = 0$.*

Proof. Combining Theorem 1.23 and Remark 1.25 we conclude that there exists a classifying map $f: S \rightarrow B\text{Spin}(n)$ for E , where n is the rank of E . From Example 1.14 (ii) it follows that $\pi_k(B\text{Spin}(n)) = 0$ for $k \geq 2$. Since S is two dimensional the map f is homotopic to a constant ([Hatcher, Lemma 4.6]), therefore $E \cong f^*\xi_{\text{Spin}}$ is trivial. $\square$